

Title: Squeak and Rattle Shaker Tests for Components or Systems

A INDEX

A Index
 B Scope
 C Object
 D Risk Assessment
 E Review
 F Related Documents
 G Equipment
 H Procedure
 I Sample Size
 J Test Failures
 K Data Generated and Reporting
 Z Change History
 Appendix A - Noise Rating Curves and Table for NR35
 Appendix B - Road Surfaces
 Appendix C - Random Input Profile
 Appendix D - Typical Test Rig Set Up

B SCOPE

This procedure applies to all Jaguar and Land Rover (JLR) components and systems on both prototype and production parts, for the purpose of squeak and rattle (S&R) assessments and static load application.

C OBJECT

The object of this procedure is to assess the following:

1 Static Test:

To assess the solidity of the system when used by the customer, by applying a load at agreed points, measuring the deflection and assessing any resultant noise.

2 Dynamic Test

To assess the Squeak & Rattle performance of the system by subjecting it to vibrations as it would experience in a vehicle and assessing any resultant unwanted / unacceptable noise.

D RISK ASSESSMENT

A valid risk assessment must be in place before this test is conducted.

E REVIEW

This document must be reviewed at least every three years from the date of the last review or update.

This document is approved for use by Jaguar Land Rover Limited	
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Title: Squeak and Rattle Shaker Tests for Components or Systems

E **RELATED DOCUMENTS**

- TPJLR 00.001 Environmental Four Poster and Proving Ground Squeak and Rattle Test
- TPJLR.00.201 Vehicle and Attribute Customer Rating System
- TPJLR.00.230 Contact Point and Material Pair Compatibility Analysis
- TPJLR.01.101 Instrument panel & glove compartment stiffness and static loads
- TPJLR.01.149 Centre console stiffness and static loads
- TPJLR.01.258 Material Compatibility

G **EQUIPMENT**

For guidance only, this may vary. Any variation to what is stated within this document must be agreed with JLR Vehicle NVH TS / SME.

Force Measuring Gauge:

- Hand held force measuring gauge capable of measuring up to 20 N, with a 10 mm diameter probe attached, manufactured from PTFE, unless otherwise agreed.
- Probe to be of appropriate material so as not to induce any damage to the surface of the test sample when a load is applied.

Microphone:

- Free field type (Class 1) ½ in (e.g. PCB 377B02)
- Sensitivity: 50 mV/Pa or higher
- Dynamic range: 144 dB or higher
- Frequency range: 10 Hz – 20 kHz (± 1dB)
- The microphone will be calibrated at least once per day using a class 1 calibrator.

Calibrator:

- Type: 94 dB; 1 kHz; Class 1 (e.g. RION NC-74)

Shaker Rig & Control System:

- Frequency range 3 Hz to 400 Hz.
- The stroke must be at least 25 mm peak to peak.
- The operating noise of the shaker system including the frame or part body shell minus the test component shall not exceed NR35 (Noise Rating curve, ISO 1996), see Appendix A.

Shaker Facility:

- The background noise, within the room, in which the shaker is located must not exceed NR25 above.

NOTE: When using calibrated instruments, ensure that the calibration is in date and that the certificate is available.

Title: Squeak and Rattle Shaker Tests for Components or Systems

H PROCEDURE

1 Set-up for Static & Shaker Test:

- 1.1 The component or assembly to be tested must be attached to a sufficiently stiff frame or part body shell in the same orientation as it is within the vehicle, using the same attachment / fixing points and methods.
- 1.2 If any environmental conditions have been requested by JLR, the assembly must be conditioned at that temperature for a minimum of 4 hours prior to test.
- 1.3 Ambient temperature for the purpose of this test must be $+21^{\circ}\text{C} \pm 3^{\circ}\text{C}$.
- 1.4 Test as per CoC Test Specification, e.g. IP & CC TPJLR. 01.149. If no specification exists then agree the specification with JLR Vehicle NVH TS / SME.

2 **Static (Load Deflection and Noise) Test:**

The detail of this test must be as per CoC TPJLR's, to be agreed with JLR Vehicle NVH TS / SME, depending on the system or component being tested.

- 2.1 The component or assembly to be tested must be set-up as instructed in section H.
- 2.2 A subjective test must initially be run, and any noise issues found, must be rated in accordance with the Vehicle and Attribute Customer Rating System (VER), as specified in TPJLR.00.201.
- 2.3 If no noise issues are found during the subjective test, no further action is required.
- 2.4 If a noise issue is heard and rated as VER 7 or lower, root cause investigation of the source is required, and a concern report must be raised to track the resolution (e.g. AIMS).
- 2.5 If sound measuring equipment and suitable software is available a further objective test may be conducted. See section H, sub-section 4.

3 **Shaker Test:**

- 3.1 The component or assembly to be tested must be set-up as specified in section H.
- 3.2 Three types of drive signal for the shaker rig are available:

Agree with JLR Vehicle NVH TS / SME which option to use.

- A. **Time History** - This is real time data measured on a vehicle at the mounting point(s) of the component or system
- B. **PSD (Power Spectral Density)** - PSD is derived from the Time History data.
- C. **Random Vibration Profile** - A generic input signal which is designed to cover all vehicle inputs if neither Real Time History nor Power Spectral Density data are available.

This input profile is given in Appendix C

Title: Squeak and Rattle Shaker Tests for Components or Systems

- 3.3 The continuous duration of the test must be a minimum of 15 minutes.
- 3.4 A subjective test must initially be run, and any noise issues found, must be rated in accordance to the Vehicle and Attribute Customer Rating System (VER), as specified in TPJLR.00.201
- 3.5 If no noise issues are found during the subjective test, no further action is required.
- 3.6 If a noise issue is heard and rated as VER 7 or lower, root cause investigation of the source is required, and a concern report must be raised to track resolution (e.g. AIMS).
- 3.7 If sound measuring equipment and suitable software are available, a further objective test may be conducted. See section H, sub-section 4.

4 Noise Recording

- 4.1 Various calibrated software packages exist which are suitable for recording and evaluating noise generated – these can be used where available.
- 4.2 JLR Vehicle NVH TS / SME will agree the objective numerical value which gives an acceptable condition with regards to squeaks & rattles, and what constitutes a fail condition.
- 4.3 Noise recordings must be supplied to JLR who will agree that the numerical rating applied, with regards to squeaks & rattles, has achieved an acceptable or fail condition.
- 4.5 The noise recordings supplied to JLR must include a sound file containing the calibration tone measured using the equipment during the calibration stage.
- 4.6 The sound measurements supplied must be recorded on the same range that was used during the calibration process.
- 4.7 The position of the microphone must be agreed with JLR Vehicle NVH TS / SME.

I SAMPLE SIZE

- 1 The sample size and frequency of testing must be agreed between JLR Vehicle NVH TS / SME, and the supplier for the various build phases, e.g. VB, TT, PP, SOP
- 2 Results must be documented in a test report and supplied to all appropriate JLR departments.

J TEST FAILURES

- 1 Any component or system that records a fail condition following the objective dynamic test as specified above must not be supplied to JLR until the following actions have been carried out.
- 2 The identification number of the component or system must be recorded.
- 3 Root cause of the issue must be established, recorded and documented, e.g. video, photo, etc.
- 4 Corrective action must be undertaken where applicable and recorded.
- 5 The component or system must successfully complete a re-test to this procedure before being supplied to JLR.

Title: Squeak and Rattle Shaker Tests for Components or Systems

K DATA GENERATED AND REPORTING

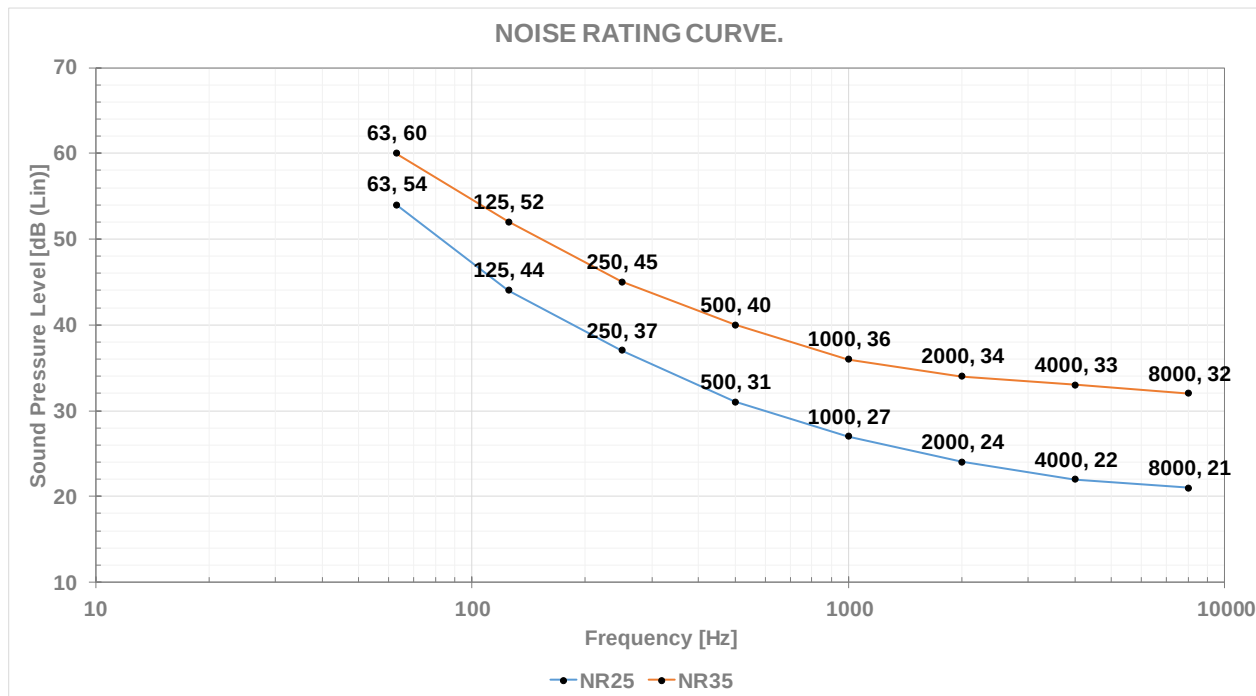
- 1 The following data must be recorded:
 - Date and time of test
 - Test facility name and location
 - Serial numbers of all the equipment used during the test
 - Calibration due date and when the equipment was last calibrated
 - Person conducting test
 - Test Data
 - All test issues must be recorded
 - All test root causes must be recorded
 - AIMS numbers, (if raised)
 - Number of test runs
 - Other relevant info as required
- 2 On completion of the test, a test report must be created and distributed to all relevant parties
- 3 The test report must be archived in accordance with current corporate records retention policy

Z CHANGE HISTORY

Issue	Date	Description of Change
1	25 May 2011	First issue
2	24 Oct 2018	General revision to procedure Clarification of the test details and revision to reporting format Update of test load figure and text Updates to test procedure text and appendices Reformatted to latest TPJLR template

Title: Squeak and Rattle Shaker Tests for Components or Systems

APPENDIX A: NOISE RATING CURVE NR35.



When conducting the test to rate the room for NR 25 compliance, the following must be adhered to:

1. All the high pass and low pass filters must be switched off in the complete measurement equipment chain during the rating of the room.
2. The whole room must comply with NR 25, including other noise sources (e.g. door, window, ventilation system, etc.) during the measurements being conducted.
3. Once the room complies to NR25, operate the shaker and retest the room. The room with the shaker operating must not exceed NR35 during the measurements being conducted.

Title: Squeak and Rattle Shaker Tests for Components or Systems

APPENDIX B: TEST TRACK SURFACES – To be recorded as drive files

1. Drive Signal Test Surfaces;

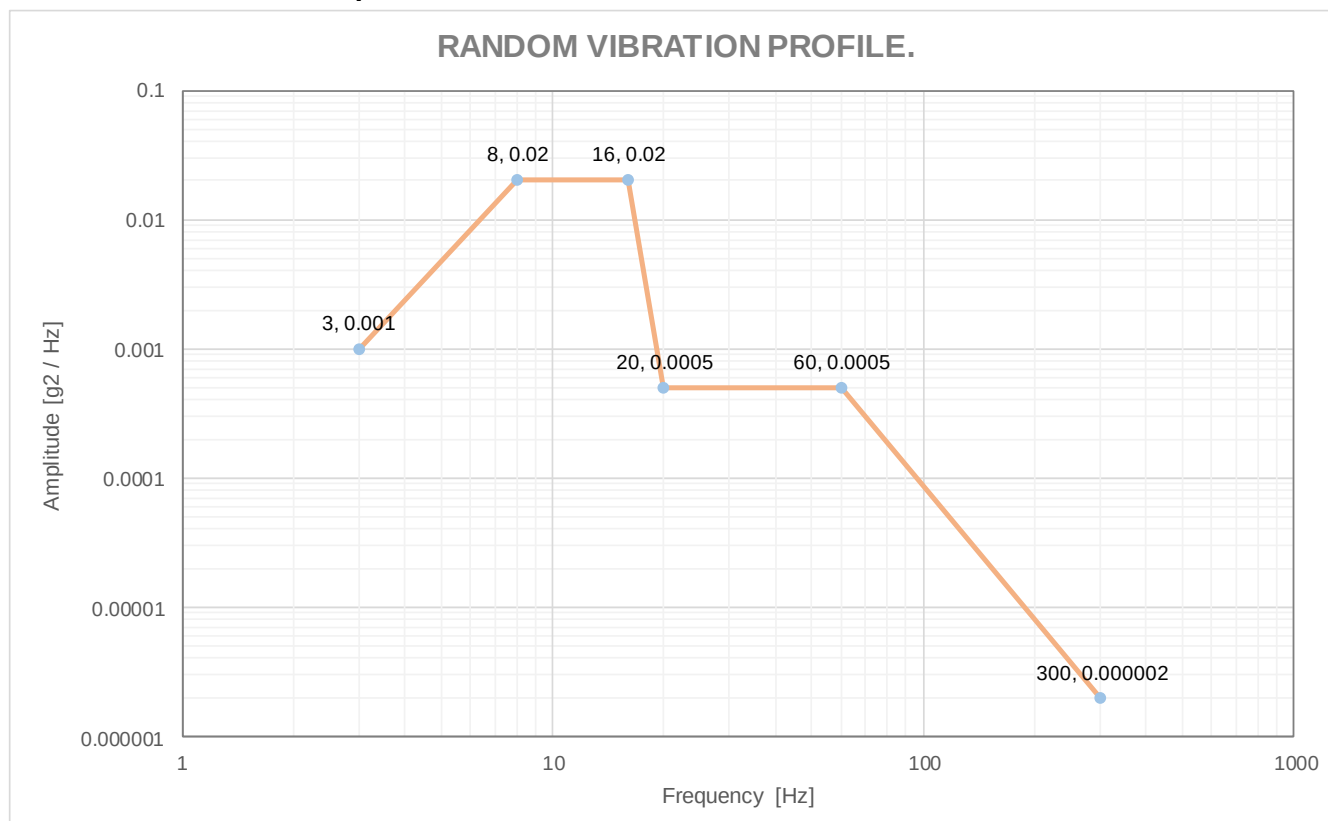
The shaker rig must have the following drive signals available:

- A. The vehicle manufacturing plant S&R test track.
- B. Gaydon PG roads;
 - W - Rough cobbled, river bed and broken slabs at 10 mph
 - V - Rough road at 10 mph
 - X - Rough road at 10 mph
 - Y - Raised bumps LHS of road (tree roots) at 10 mph
 - Z - Raised manhole covers road at 10 mph
 - HIST - Course Chip road at 10 mph
 - Torsion Twist - if appropriate

The above sequence must be repeated for 15 mins minimum as the assessment test.

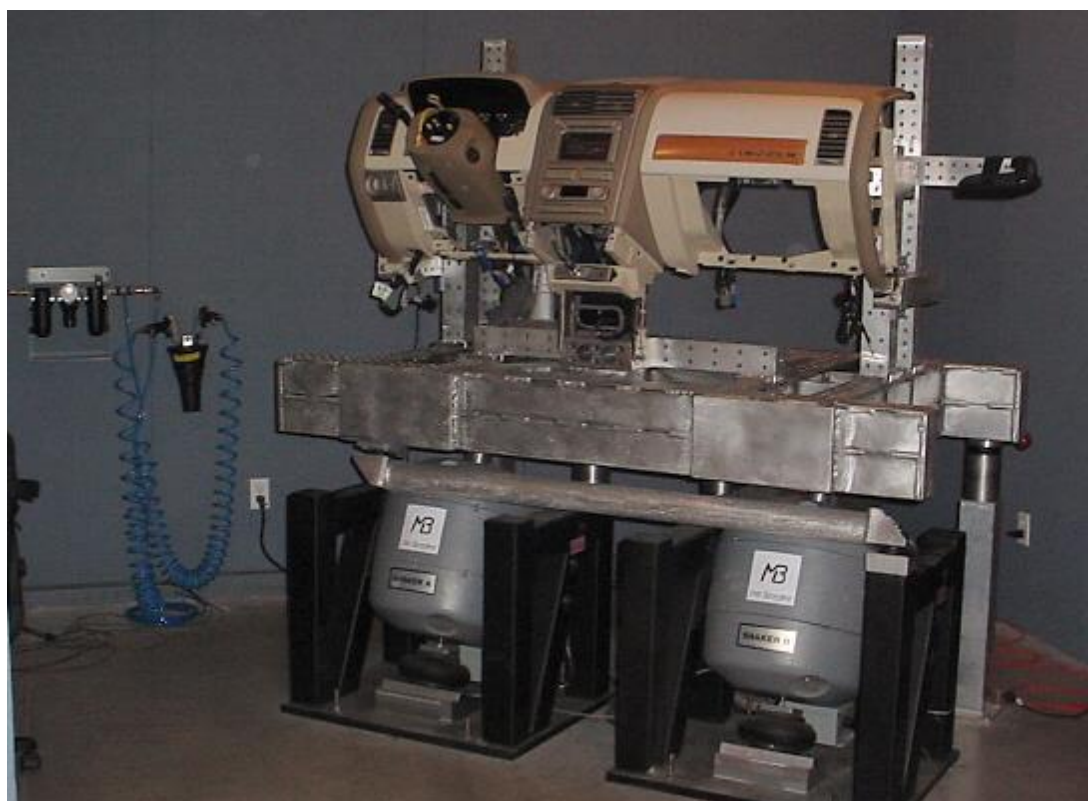
Other surfaces can be added to the library and used to aid root cause work as needed, including sine sweep.

APPENDIX C: Random Input Profile



Title: Squeak and Rattle Shaker Tests for Components or Systems

APPENDIX D: Typical Rig Set-Up



Title: Squeak and Rattle Shaker Tests for Components or Systems

